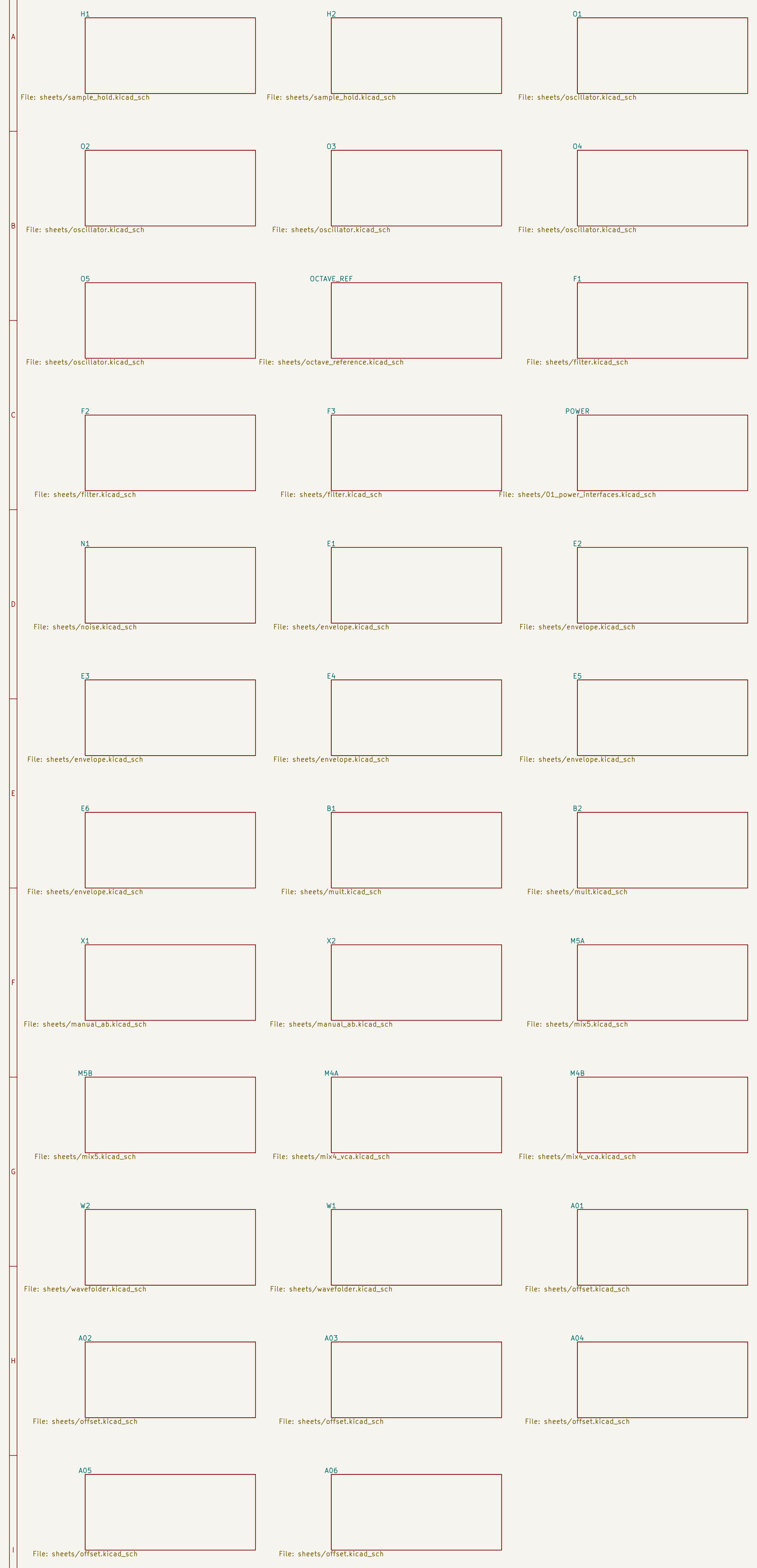
H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

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File: zudo-osc-hole-field.kicad_sch

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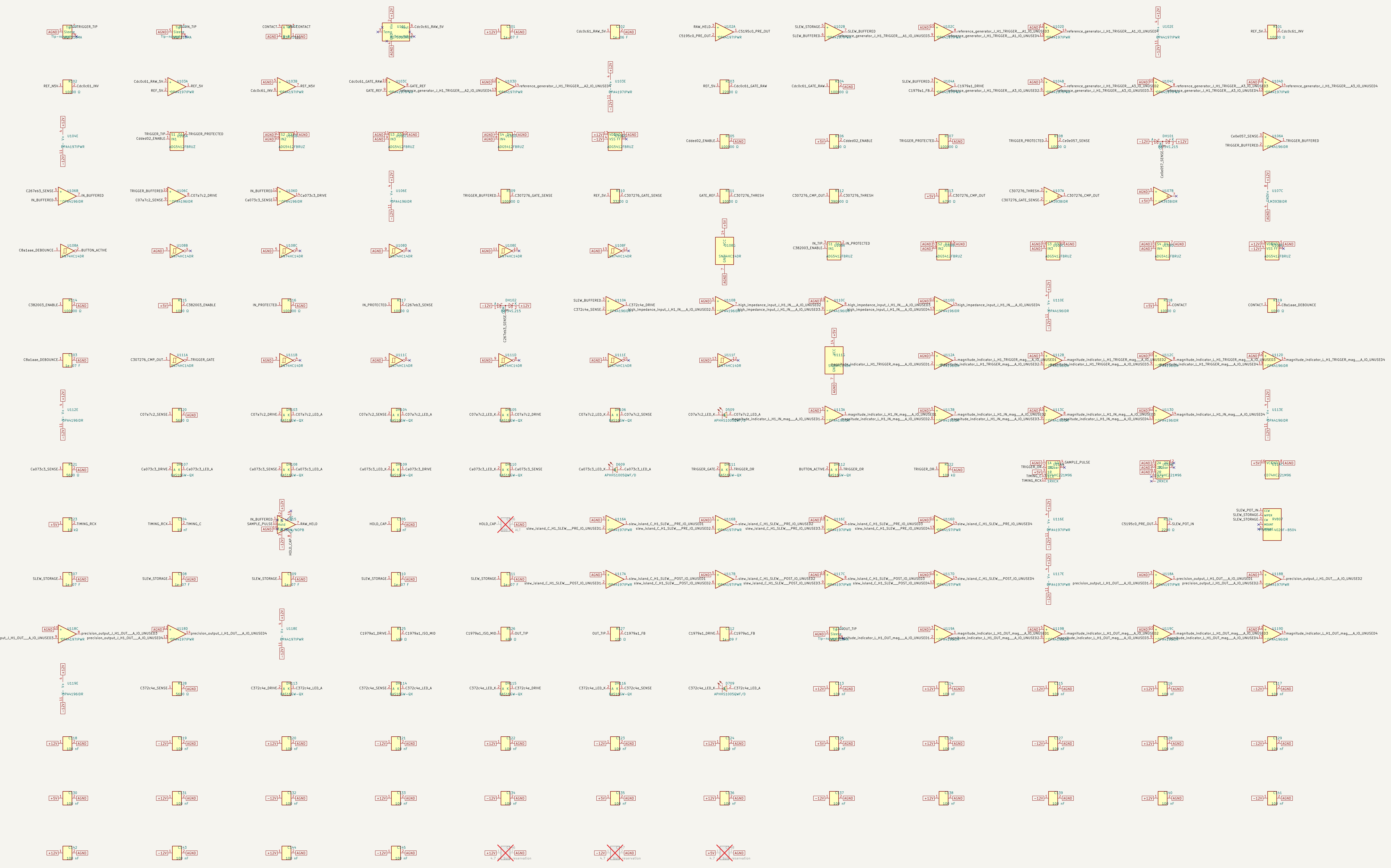
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/36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RETURN_TIP
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Cdc0c61_INV
REF_5V
REF_N5V
RAW_HELD
C5195c0_PRE_OUT
SLEW_STORAGE
SLEW_BUFFERED
Cdc0c61_GATE_RAW
GATE_REF
C1979a1_DRIVE
C1979a1_FB
TRIGGER_PROTECTED
Cdded02_ENABLE
Ce0e057_SENSE
TRIGGER_BUFFERED
C267eb3_SENSE
IN_BUFFERED
C07a7c2_DRIVE
C07a7c2_SENSE
Ca073c3_DRIVE
Ca073c3_SENSE
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
C8a1aae_DEBOUNCE
BUTTON_ACTIVE
IN_TIP
IN_PROTECTED
C382003_ENABLE
C372c4e_DRIVE
C372c4e_SENSE
TRIGGER_GATE
C07a7c2_LED_A
C07a7c2_LED_K
Ca073c3_LED_A
Ca073c3_LED_K
TRIGGER_OR
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
SLEW_POT_IN
C1979a1_ISO_MID
OUT_TIP
C372c4e_LED_A
C372c4e_LED_K
AGND
+12V
-12V
+5V
OPA4197IPWR
OPA4196IDR
SN74HC14DR
ADG5412FBRUZ
LM393BIDR
BAT54S,215
BAS16GW-QX
APHHS1005QWF/D
CD74HC221M96
PT4028F-B504
Tip-normal

Sheet: /H1/
File: sample_hold.kicad_sch

Title:

Size: A1
Date:
KiCad E.D.A. 10.0.6
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Id: 2/36

AGND
+12V
-12V
+5V
OPA4197IPWR
OPA4196IDR
SN74HC14DR
ADG5412FBRUZ
BAS16W-QX
APHHS1005QWF/D
BAT54S,215
LM393BIDR
CD74HC221M96
RAW_HELD
SLEW_STORAGE
SLEW_BUFFERED
TRIGGER_PROTECTED
TRIGGER_BUFFERED
IN_BUFFERED
IN_PROTECTED
SAMPLE_PULSE
HOLD_CAP
TIMING_RCX
TIMING_C
OUT_TIP
TRIGGER_GATE
TRIGGER_OR
BUTTON_ACTIVE
CONTACT
REF_5V
REF_N5V

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Size: A1 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 3/36

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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 4/36

Sheet: /02/
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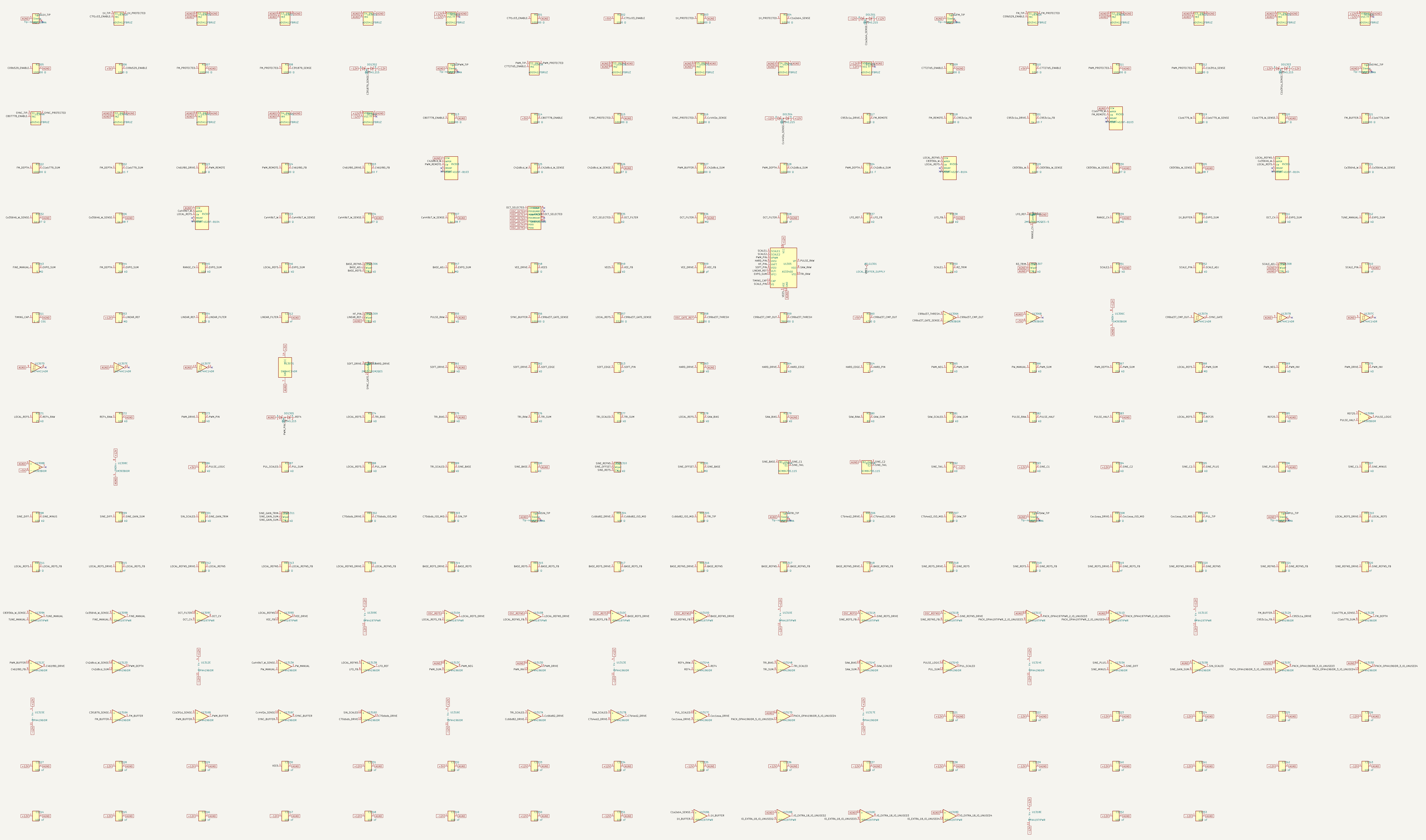
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/36



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KiCad E.D.A. 10.0.6 | Id: 6/36

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KiCad E.D.A. 10.0.6 Id: 7/36

Sheet: /05/
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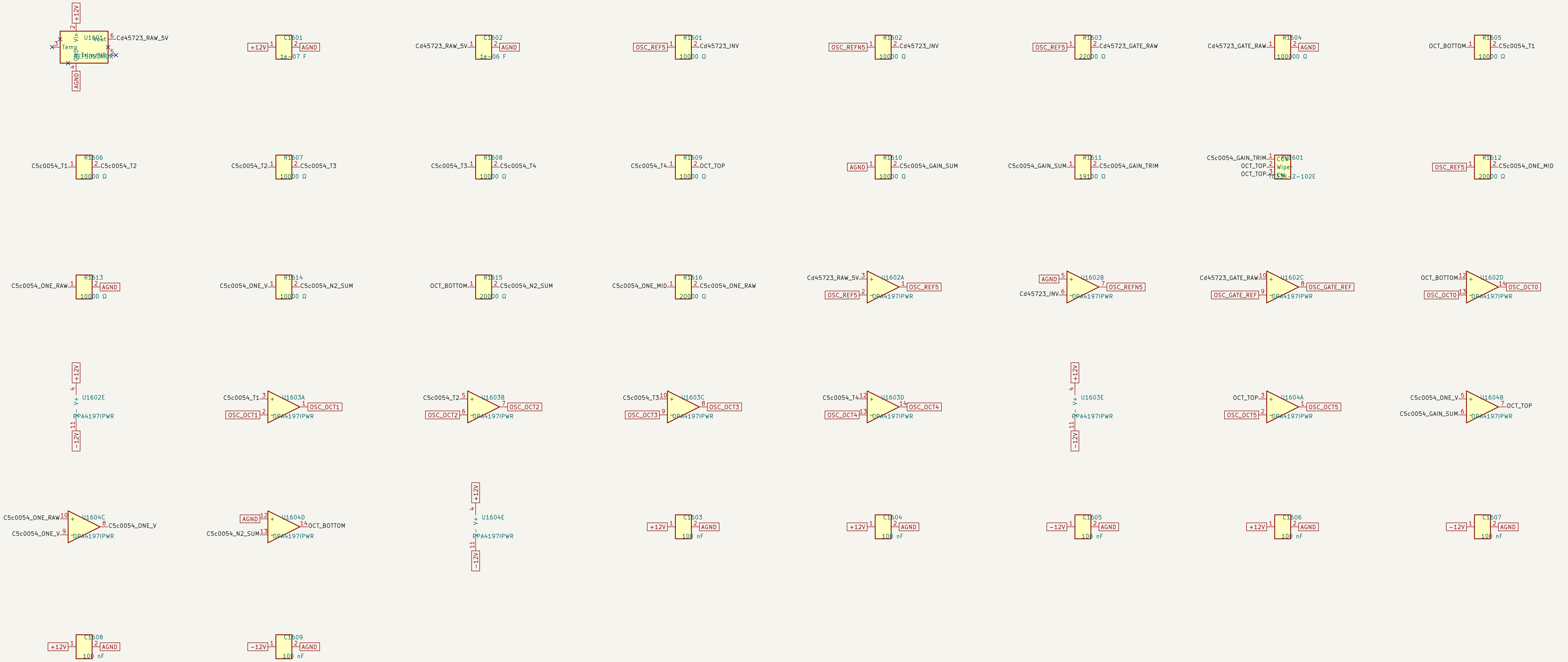
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 8/36


+12V
U1601
Vin
Vout
Temp
GND
Trim/NR
REF5050AIDR
Cd45723_RAW_5V
AGND
+12V
C1601
1e-07 F
AGND
Cd45723_RAW_5V
C1602
1e-06 F
AGND
OSC_REF5
R1601
10000 Ω
Cd45723_INV
OSC_REFN5
R1602
10000 Ω
Cd45723_INV
OSC_REF5
R1603
22000 Ω
Cd45723_GATE_RAW
Cd45723_GATE_RAW
R1604
100000 Ω
AGND
OCT_BOTTOM
R1605
10000 Ω
C5c0054_T1
C5c0054_T1
R1606
10000 Ω
C5c0054_T2
C5c0054_T2
R1607
10000 Ω
C5c0054_T3
C5c0054_T3
R1608
10000 Ω
C5c0054_T4
C5c0054_T4
R1609
10000 Ω
OCT_TOP
AGND
R1610
10000 Ω
C5c0054_GAIN_SUM
C5c0054_GAIN_SUM
R1611
19100 Ω
C5c0054_GAIN_TRIM
C5c0054_GAIN_TRIM
OCT_TOP
OCT_TOP
RV1601
CCW
Wiper
CW
TC33X-2-102E
OSC_REF5
R1612
20000 Ω
C5c0054_ONE_MID
C5c0054_ONE_RAW
R1613
10000 Ω
AGND
C5c0054_ONE_V
R1614
10000 Ω
C5c0054_N2_SUM
OCT_BOTTOM
R1615
20000 Ω
C5c0054_N2_SUM
C5c0054_ONE_MID
R1616
20000 Ω
C5c0054_ONE_RAW
Cd45723_RAW_5V
OSC_REF5
U1602A
OPA4197IPWR
OSC_REF5
AGND
Cd45723_INV
U1602B
OPA4197IPWR
OSC_REFN5
Cd45723_GATE_RAW
OSC_GATE_REF
U1602C
OPA4197IPWR
OSC_GATE_REF
OCT_BOTTOM
OSC_OCT0
U1602D
OPA4197IPWR
OSC_OCT0
+12V
V+
V-
U1602E
OPA4197IPWR
-12V
C5c0054_T1
OSC_OCT1
U1603A
OPA4197IPWR
OSC_OCT1
C5c0054_T2
OSC_OCT2
U1603B
OPA4197IPWR
OSC_OCT2
C5c0054_T3
OSC_OCT3
U1603C
OPA4197IPWR
OSC_OCT3
C5c0054_T4
OSC_OCT4
U1603D
OPA4197IPWR
OSC_OCT4
+12V
V+
V-
U1603E
OPA4197IPWR
-12V
OCT_TOP
OSC_OCT5
U1604A
OPA4197IPWR
OSC_OCT5
C5c0054_ONE_V
C5c0054_GAIN_SUM
U1604B
OPA4197IPWR
OCT_TOP
C5c0054_ONE_RAW
C5c0054_ONE_V
U1604C
OPA4197IPWR
C5c0054_ONE_V
AGND
C5c0054_N2_SUM
U1604D
OPA4197IPWR
OCT_BOTTOM
+12V
V+
V-
U1604E
OPA4197IPWR
-12V
+12V
C1603
100 nF
AGND
+12V
C1604
100 nF
AGND
-12V
C1605
100 nF
AGND
+12V
C1606
100 nF
AGND
-12V
C1607
100 nF
AGND
+12V
C1608
100 nF
AGND
-12V
C1609
100 nF
AGND

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KiCad E.D.A. 10.0.6
Rev:
Id: 9/36

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Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 10/36

Sheet: /F2/
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Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 11/36

Sheet: /F3/
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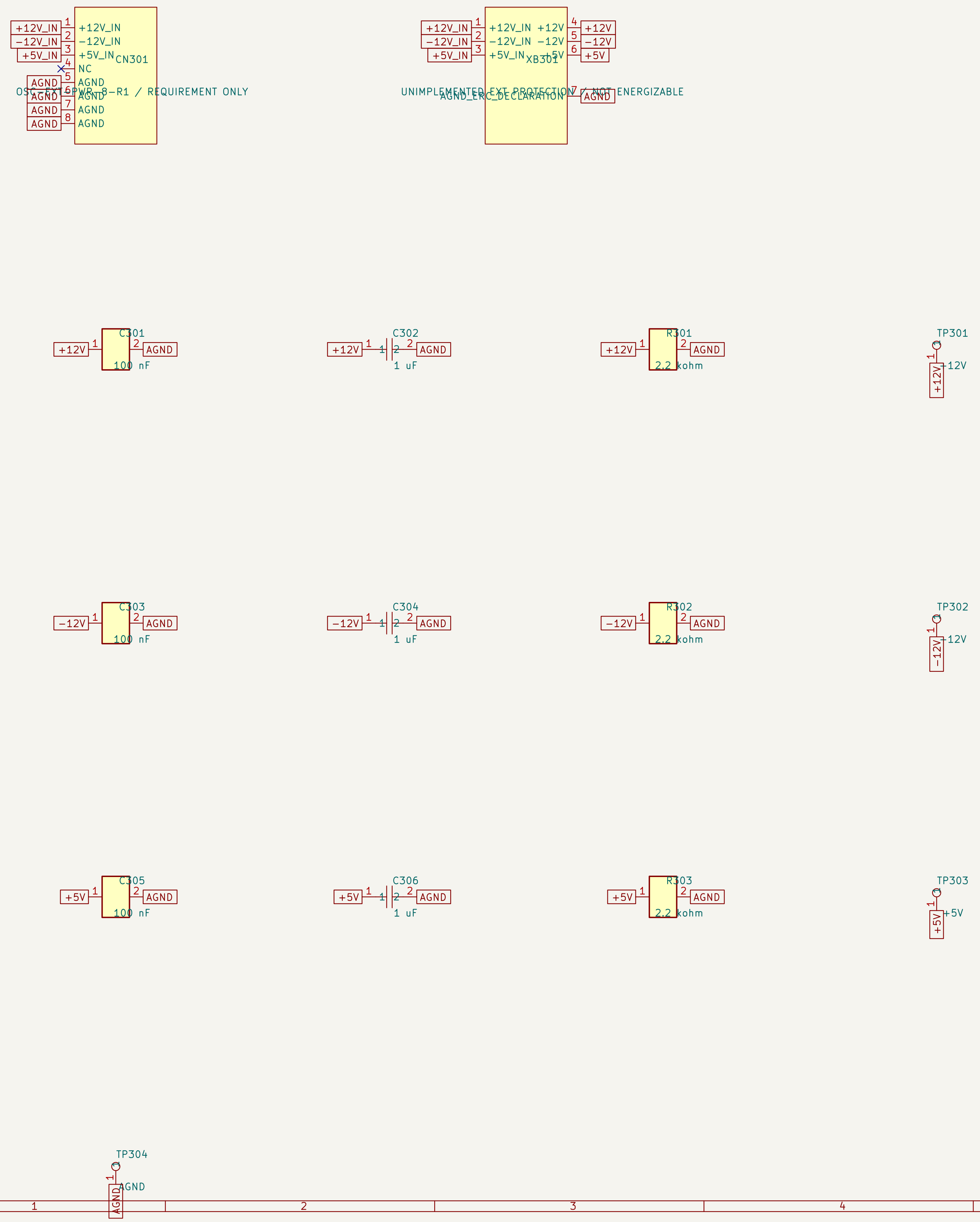
Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 12/36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+12V_IN
-12V_IN
+5V_IN
NC
AGND
AGND
AGND
AGND
OSC-EXT-PWR-8-R1 / REQUIREMENT ONLY
XB301
+12V_IN
-12V_IN
+5V_IN
+12V
-12V
+5V
AGND_ERC_DECLARATION
AGND
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
C301
100 nF
C302
1 uF
R301
2.2 kohm
TP301
+12V
C303
100 nF
C304
1 uF
R302
2.2 kohm
TP302
-12V
C305
100 nF
C306
1 uF
R303
2.2 kohm
TP303
+5V
TP304
AGND

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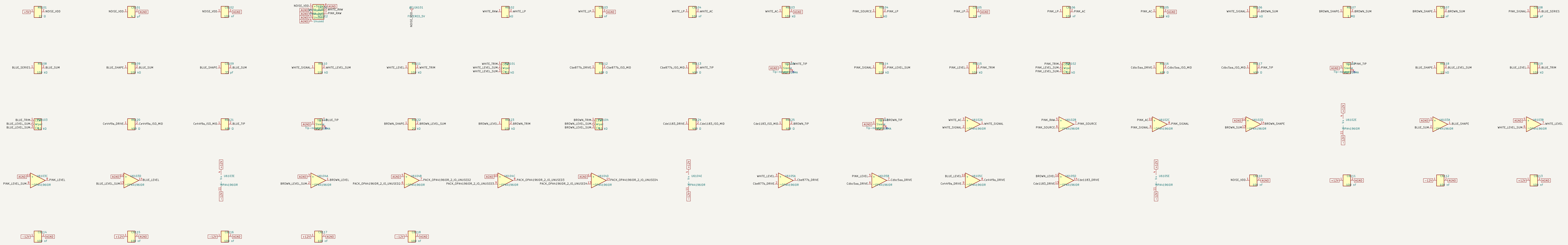
File: 01_power_interfaces.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 13/36


R6101 22 Ω +5V NOISE_VDD
C6101 4.7 µF NOISE_VDD AGND
C6102 100 nF NOISE_VDD AGND
U6101 NOISE2 NOISE_VDD +5V 0V AGND WHITE_OUT WHITE_RAW PINK_OUT PINK_RAW Unused Unused AGND
#FLG6101 FILTERED_5V NOISE_VDD
R6102 1 kΩ WHITE_RAW WHITE_LP
C6103 10 nF WHITE_LP AGND
C6104 100 nF WHITE_LP WHITE_AC
R6103 100 kΩ WHITE_AC AGND
R6104 1 kΩ PINK_SOURCE PINK_LP
C6105 10 nF PINK_LP AGND
C6106 100 nF PINK_LP PINK_AC
R6105 100 kΩ PINK_AC AGND
R6106 100 kΩ WHITE_SIGNAL BROWN_SUM
R6107 1 MΩ BROWN_SHAPE BROWN_SUM
C6107 10 nF BROWN_SHAPE BROWN_SUM
C6108 100 pF PINK_SIGNAL BLUE_SERIES
R6108 100 kΩ BLUE_SERIES BLUE_SUM
R6109 100 kΩ BLUE_SHAPE BLUE_SUM
C6109 33 pF BLUE_SHAPE BLUE_SUM
R6110 100 kΩ WHITE_SIGNAL WHITE_LEVEL_SUM
R6111 100 kΩ WHITE_LEVEL WHITE_TRIM
RV6101 10 kΩ CCW Wiper CW WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM
R6112 499 Ω Cbe877b_DRIVE Cbe877b_ISO_MID
R6113 499 Ω Cbe877b_ISO_MID WHITE_TIP
Tip Sleeve Tip-normal TN AGND WHITE_TIP
R6114 100 kΩ PINK_SIGNAL PINK_LEVEL_SUM
R6115 100 kΩ PINK_LEVEL PINK_TRIM
RV6102 10 kΩ CCW Wiper CW PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM
R6116 499 Ω Cdbc5aa_DRIVE Cdbc5aa_ISO_MID
R6117 499 Ω Cdbc5aa_ISO_MID PINK_TIP
Tip Sleeve Tip-normal TN AGND PINK_TIP
R6118 10 kΩ BLUE_SHAPE BLUE_LEVEL_SUM
R6119 100 kΩ BLUE_LEVEL BLUE_TRIM
RV6103 10 kΩ CCW Wiper CW BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM
R6120 499 Ω Ce44f9a_DRIVE Ce44f9a_ISO_MID
R6121 499 Ω Ce44f9a_ISO_MID BLUE_TIP
Tip Sleeve Tip-normal TN AGND BLUE_TIP
R6122 20 kΩ BROWN_SHAPE BROWN_LEVEL_SUM
R6123 100 kΩ BROWN_LEVEL BROWN_TRIM
RV6104 10 kΩ CCW Wiper CW BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM
R6124 499 Ω Cde1183_DRIVE Cde1183_ISO_MID
R6125 499 Ω Cde1183_ISO_MID BROWN_TIP
Tip Sleeve Tip-normal TN AGND BROWN_TIP
U6102A OPA4196IDR WHITE_AC WHITE_SIGNAL WHITE_SIGNAL
U6102B OPA4196IDR PINK_RAW PINK_SOURCE PINK_SOURCE
U6102C OPA4196IDR PINK_AC PINK_SIGNAL PINK_SIGNAL
U6102D OPA4196IDR AGND BROWN_SUM BROWN_SHAPE
U6102E OPA4196IDR +12V -12V V+ V-
U6103A OPA4196IDR AGND BLUE_SUM BLUE_SHAPE
U6103B OPA4196IDR AGND WHITE_LEVEL_SUM WHITE_LEVEL
U6103C OPA4196IDR AGND PINK_LEVEL_SUM PINK_LEVEL
U6103D OPA4196IDR AGND BLUE_LEVEL_SUM BLUE_LEVEL
U6103E OPA4196IDR +12V -12V V+ V-
U6104A OPA4196IDR AGND BROWN_LEVEL_SUM BROWN_LEVEL
U6104B OPA4196IDR AGND PACK_OPA4196IDR_2_IO_UNUSED2 PACK_OPA4196IDR_2_IO_UNUSED2
U6104C OPA4196IDR AGND PACK_OPA4196IDR_2_IO_UNUSED3 PACK_OPA4196IDR_2_IO_UNUSED3
U6104D OPA4196IDR AGND PACK_OPA4196IDR_2_IO_UNUSED4 PACK_OPA4196IDR_2_IO_UNUSED4
U6104E OPA4196IDR +12V -12V V+ V-
U6105A OPA4196IDR WHITE_LEVEL Cbe877b_DRIVE Cbe877b_DRIVE
U6105B OPA4196IDR PINK_LEVEL Cdbc5aa_DRIVE Cdbc5aa_DRIVE
U6105C OPA4196IDR BLUE_LEVEL Ce44f9a_DRIVE Ce44f9a_DRIVE
U6105D OPA4196IDR BROWN_LEVEL Cde1183_DRIVE Cde1183_DRIVE
U6105E OPA4196IDR +12V -12V V+ V-
C6110 100 nF NOISE_VDD AGND
C6111 100 nF +12V AGND
C6112 100 nF -12V AGND
C6113 100 nF +12V AGND
C6114 100 nF -12V AGND
C6115 100 nF +12V AGND
C6116 100 nF -12V AGND
C6117 100 nF +12V AGND
C6118 100 nF -12V AGND

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Title:

Size: A0 Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 14/36

Sheet: /E1/
File: envelope.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 15/36

Sheet: /E2/
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Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 16/36

Sheet: /E3/
File: envelope.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 17/36

Sheet: /E4/
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Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 18/36

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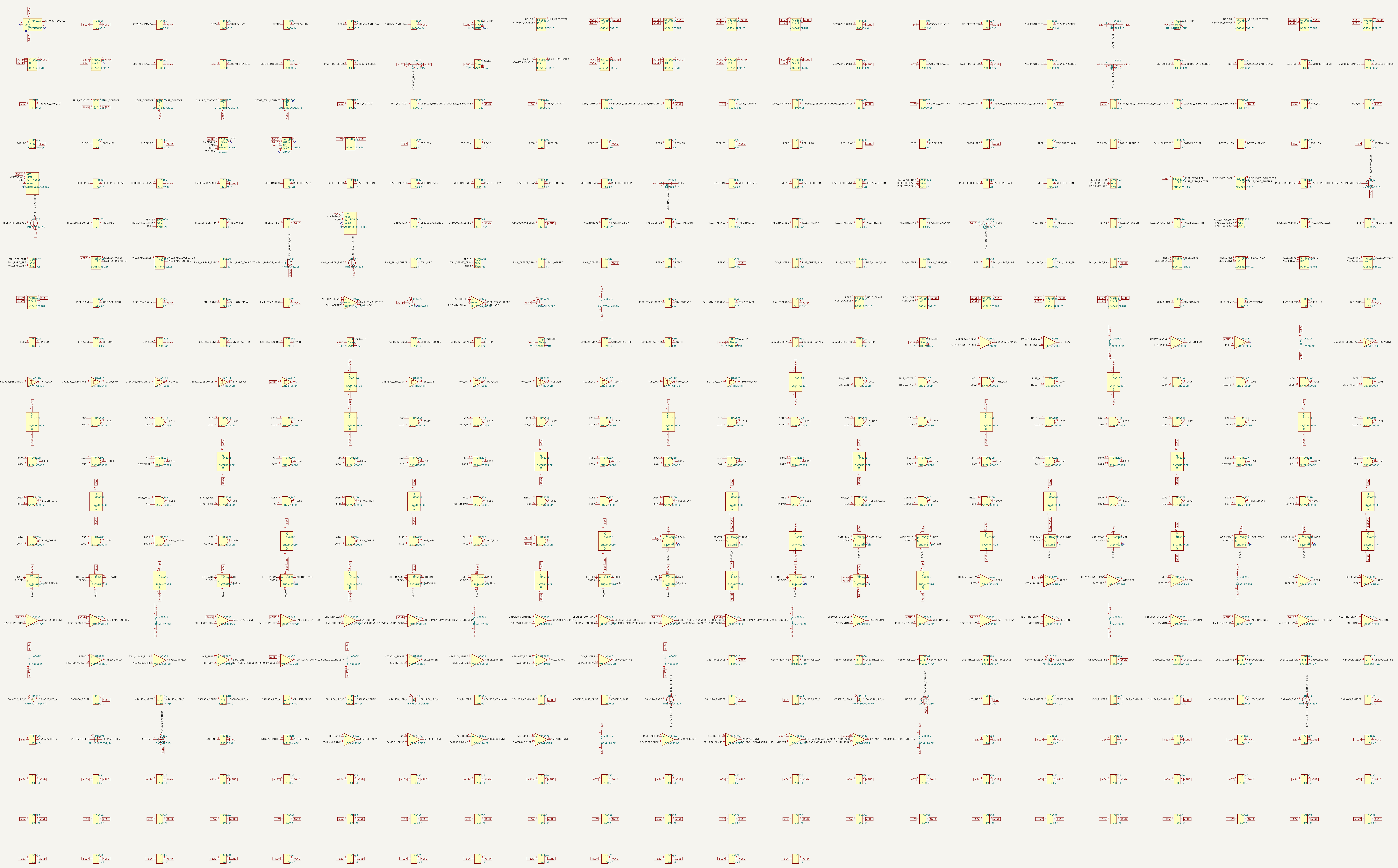
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 19/36



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Size: A0 Date: **Rev:**
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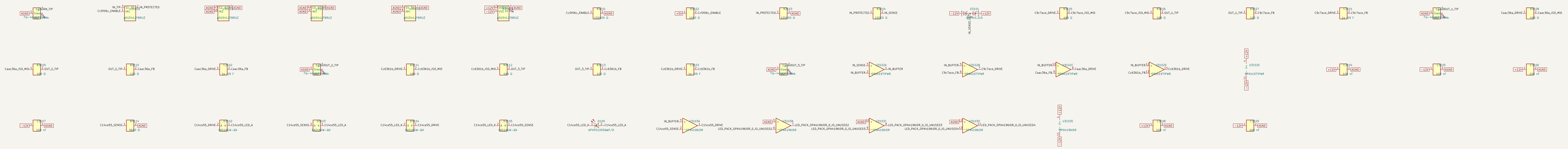
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Tip
Sleeve
IN_TIP
Tip-normal
WQP518MA
IN_TIP
Cc5f06c_ENABLE
IN_PROTECTED
S1
D1
IN1
U3101A
ADG5412FBRUZ
S2
D2
IN2
U3101B
ADG5412FBRUZ
S3
D3
IN3
U3101C
ADG5412FBRUZ
S4
D4
IN4
U3101D
ADG5412FBRUZ
+12V
-12V
VDD
VSS
U3101E
ADG5412FBRUZ
R3101
100000 Ω
+5V
R3102
1000 Ω
R3103
100000 Ω
IN_PROTECTED
R3104
10000 Ω
IN_SENSE
D3101
BAT54S,215
C9c7ace_DRIVE
R3105
499 Ω
C9c7ace_ISO_MID
R3106
499 Ω
OUT_1_TIP
R3107
100 Ω
C9c7ace_FB
C3101
1e-09 F
OUT_1_TIP
Caac36a_DRIVE
R3108
499 Ω
Caac36a_ISO_MID
R3109
499 Ω
OUT_2_TIP
R3110
100 Ω
Caac36a_FB
C3102
1e-09 F
OUT_2_TIP
Cc6361b_DRIVE
R3111
499 Ω
Cc6361b_ISO_MID
R3112
499 Ω
OUT_3_TIP
R3113
100 Ω
Cc6361b_FB
C3103
1e-09 F
OUT_3_TIP
IN_SENSE
IN_BUFFER
U3102A
OPA4197IPWR
U3102B
OPA4197IPWR
U3102C
OPA4197IPWR
U3102D
OPA4197IPWR
V+
V-
U3102E
OPA4197IPWR
C3104
100 nF
C3105
100 nF
C3106
100 nF
C3107
100 nF
C14ce55_SENSE
R3114
5600 Ω
C14ce55_DRIVE
D3102
BAS16GW-QX
C14ce55_LED_A
D3103
BAS16GW-QX
C14ce55_LED_K
D3104
BAS16GW-QX
D3105
BAS16GW-QX
D109
APHHS1005QWF/D
U3103A
OPA4196IDR
U3103B
OPA4196IDR
LED_PACK_OPA4196IDR_0_IO_UNUSED2
U3103C
OPA4196IDR
LED_PACK_OPA4196IDR_0_IO_UNUSED3
U3103D
OPA4196IDR
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3103E
OPA4196IDR
C3108
100 nF
C3109
100 nF

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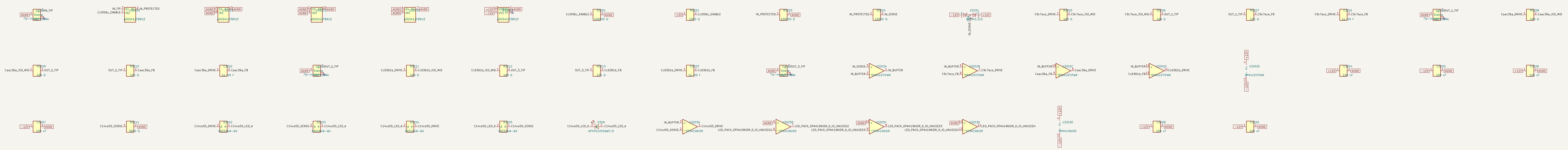
Title:

Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 21/36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Sleeve
Tip-normal
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WQP518MA
IN_TIP
Cc5f06c_ENABLE
S1 D1
IN1
U3201A
IN_PROTECTED
ADG5412FBRUZ
AGND
S2 D2
IN2
U3201B
ADG5412FBRUZ
S3 D3
IN3
U3201C
ADG5412FBRUZ
S4 D4
IN4
U3201D
ADG5412FBRUZ
+12V
-12V
VDD GND
VSS FF
U3201E
ADG5412FBRUZ
R3201
Cc5f06c_ENABLE
100000 Ω
R3202
+5V
1000 Ω
R3203
IN_PROTECTED
100000 Ω
R3204
IN_SENSE
10000 Ω
D3201
BAT54S,215
R3205
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3206
OUT_1_TIP
499 Ω
R3207
C9c7ace_FB
100 Ω
C3201
1e-09 F
R3208
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3209
OUT_2_TIP
499 Ω
R3210
Caac36a_FB
100 Ω
C3202
1e-09 F
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Cc6361b_DRIVE
Cc6361b_ISO_MID
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R3212
OUT_3_TIP
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Cc6361b_FB
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C3203
1e-09 F
U3202A
IN_BUFFER
OPA4197IPWR
U3202B
U3202C
U3202D
U3202E
V+
V-
C3204
100 nF
C3205
C3206
C3207
R3214
C14ce55_SENSE
5600 Ω
D3202
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3203
D3204
C14ce55_LED_K
D3205
D309
APHHS1005QWF/D
U3203A
OPA4196IDR
U3203B
LED_PACK_OPA4196IDR_0_IO_UNUSED2
U3203C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
U3203D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3203E
C3208
C3209

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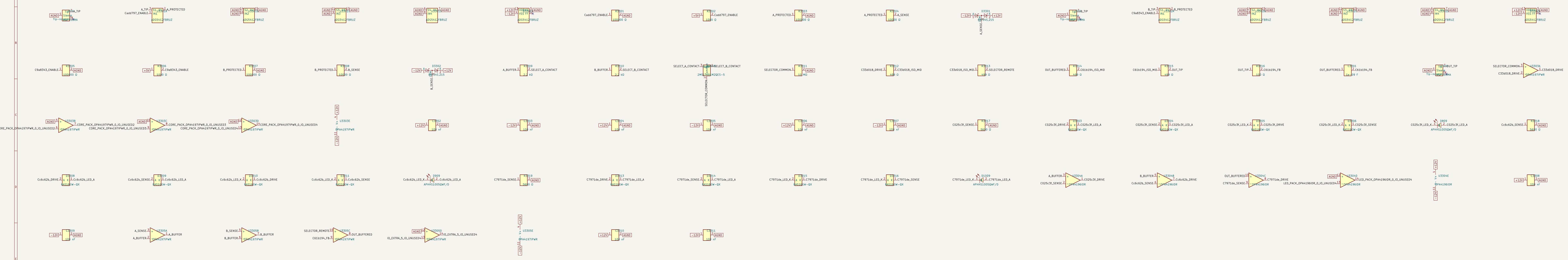
AGND
A_TIP
Tip-normal
U3301A
A_TIP
Cadd797_ENABLE
A_PROTECTED
ADG5412FBRUZ
U3301B
ADG5412FBRUZ
U3301C
ADG5412FBRUZ
U3301D
ADG5412FBRUZ
U3301E
+12V
-12V
VDD
VSS
ADG5412FBRUZ
R3301
Cadd797_ENABLE
100000 Ω
R3302
+5V
Cadd797_ENABLE
1000 Ω
R3303
A_PROTECTED
100000 Ω
R3304
A_PROTECTED
A_SENSE
10000 Ω
D3301
A_SENSE
BAT54S,215
B_TIP
Tip-normal
U3302A
B_TIP
C9a8343_ENABLE
B_PROTECTED
ADG5412FBRUZ
U3302B
ADG5412FBRUZ
U3302C
ADG5412FBRUZ
U3302D
ADG5412FBRUZ
U3302E
ADG5412FBRUZ
R3305
C9a8343_ENABLE
100000 Ω
R3306
C9a8343_ENABLE
1000 Ω
R3307
B_PROTECTED
100000 Ω
R3308
B_PROTECTED
B_SENSE
10000 Ω
D3302
B_SENSE
BAT54S,215
R3309
A_BUFFER
SELECT_A_CONTACT
2.2 kΩ
R3310
B_BUFFER
SELECT_B_CONTACT
2.2 kΩ
SELECT_A_CONTACT
SELECT_B_CONTACT
SELECTOR_COMMON
2MS1T1B1M2QES-5
R3311
SELECTOR_COMMON
10 MΩ
R3312
C33d018_DRIVE
C33d018_ISO_MID
499 Ω
R3313
C33d018_ISO_MID
SELECTOR_REMOTE
499 Ω
R3314
OUT_BUFFERED
C61b194_ISO_MID
499 Ω
R3315
C61b194_ISO_MID
OUT_TIP
499 Ω
R3316
OUT_TIP
C61b194_FB
100 Ω
C3301
OUT_BUFFERED
C61b194_FB
1e-09 F
OUT_TIP
Tip-normal
U3303A
SELECTOR_COMMON
C33d018_DRIVE
C33d018_DRIVE
OPA4197IPWR
U3303B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
OPA4197IPWR
U3303C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
OPA4197IPWR
U3303D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
OPA4197IPWR
U3303E
OPA4197IPWR
C3302
100 nF
C3303
100 nF
C3304
100 nF
C3305
100 nF
C3306
100 nF
C3307
100 nF
R3317
C025c3f_SENSE
5600 Ω
D3303
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3304
C025c3f_SENSE
C025c3f_LED_A
BAS16GW-QX
D3305
C025c3f_LED_K
C025c3f_DRIVE
BAS16GW-QX
D3306
C025c3f_LED_K
C025c3f_SENSE
BAS16GW-QX
D809
C025c3f_LED_K
C025c3f_LED_A
APHHS1005QWF/D
R3318
Cc6c62b_SENSE
5600 Ω
D3308
Cc6c62b_DRIVE
Cc6c62b_LED_A
BAS16GW-QX
D3309
Cc6c62b_SENSE
Cc6c62b_LED_A
BAS16GW-QX
D3310
Cc6c62b_LED_K
Cc6c62b_DRIVE
BAS16GW-QX
D3311
Cc6c62b_LED_K
Cc6c62b_SENSE
BAS16GW-QX
D909
Cc6c62b_LED_K
Cc6c62b_LED_A
APHHS1005QWF/D
R3319
C7971de_SENSE
5600 Ω
D3313
C7971de_DRIVE
C7971de_LED_A
BAS16GW-QX
D3314
C7971de_SENSE
C7971de_LED_A
BAS16GW-QX
D3315
C7971de_LED_K
C7971de_DRIVE
BAS16GW-QX
D3316
C7971de_LED_K
C7971de_SENSE
BAS16GW-QX
D1009
C7971de_LED_K
C7971de_LED_A
APHHS1005QWF/D
U3304A
A_BUFFER
C025c3f_SENSE
C025c3f_DRIVE
OPA4196IDR
U3304B
B_BUFFER
Cc6c62b_SENSE
Cc6c62b_DRIVE
OPA4196IDR
U3304C
OUT_BUFFERED
C7971de_SENSE
C7971de_DRIVE
OPA4196IDR
U3304D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3304E
OPA4196IDR
C3308
100 nF
C3309
100 nF
U3305A
A_SENSE
A_BUFFER
A_BUFFER
OPA4197IPWR
U3305B
B_SENSE
B_BUFFER
B_BUFFER
OPA4197IPWR
U3305C
SELECTOR_REMOTE
C61b194_FB
OUT_BUFFERED
OPA4197IPWR
U3305D
IO_EXTRA_5_IO_UNUSED4
OPA4197IPWR
U3305E
OPA4197IPWR
C3310
100 nF
C3311
100 nF
+12V
-12V
+5V
AGND

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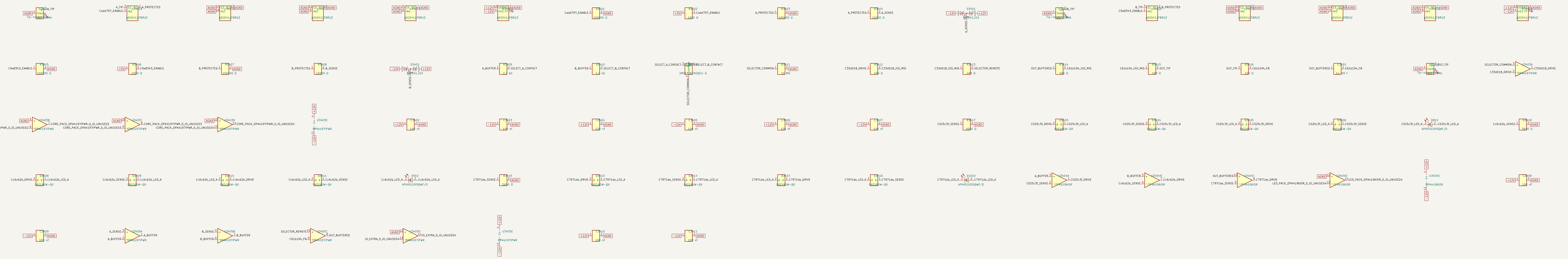
AGND S Tip Sleeve TN J810 A_TIP Tip-normal WQP518MA
A_TIP 3 S1 D1 2 A_PROTECTED Cadd797_ENABLE 1 IN1 U3401A ADG5412FBRUZ
AGND 14 S2 D2 15 AGND AGND 16 IN2 U3401B ADG5412FBRUZ
AGND 11 S3 D3 10 AGND AGND 9 IN3 U3401C ADG5412FBRUZ
AGND 6 S4 D4 7 AGND AGND 8 IN4 U3401D ADG5412FBRUZ
+12V 13 VDD GND 5 AGND −12V 4 VSS FF 12 U3401E ADG5412FBRUZ
Cadd797_ENABLE 1 R3401 2 AGND 100000 Ω
+5V 1 R3402 2 Cadd797_ENABLE 1000 Ω
A_PROTECTED 1 R3403 2 AGND 100000 Ω
A_PROTECTED 1 R3404 2 A_SENSE 10000 Ω
−12V A 1 D3401 K 2 +12V BAT54S,215 A_SENSE
AGND S Tip Sleeve TN J910 B_TIP Tip-normal WQP518MA
B_TIP 3 S1 D1 2 B_PROTECTED C9a8343_ENABLE 1 IN1 U3402A ADG5412FBRUZ
AGND 14 S2 D2 15 AGND AGND 16 IN2 U3402B ADG5412FBRUZ
AGND 11 S3 D3 10 AGND AGND 9 IN3 U3402C ADG5412FBRUZ
AGND 6 S4 D4 7 AGND AGND 8 IN4 U3402D ADG5412FBRUZ
+12V 13 VDD GND 5 AGND −12V 4 VSS FF 12 U3402E ADG5412FBRUZ
C9a8343_ENABLE 1 R3405 2 AGND 100000 Ω
+5V 1 R3406 2 C9a8343_ENABLE 1000 Ω
B_PROTECTED 1 R3407 2 AGND 100000 Ω
B_PROTECTED 1 R3408 2 B_SENSE 10000 Ω
−12V A 1 D3402 K 2 +12V BAT54S,215 B_SENSE
A_BUFFER 1 R3409 2 SELECT_A_CONTACT 2.2 kΩ
B_BUFFER 1 R3410 2 SELECT_B_CONTACT 2.2 kΩ
SELECT_A_CONTACT 3 Right Left Common 1 SELECT_B_CONTACT 2MS1T1B1M2QES-5 2 SELECTOR_COMMON
SELECTOR_COMMON 1 R3411 2 AGND 10 MΩ
C33d018_DRIVE 1 R3412 2 C33d018_ISO_MID 499 Ω
C33d018_ISO_MID 1 R3413 2 SELECTOR_REMOTE 499 Ω
OUT_BUFFERED 1 R3414 2 C61b194_ISO_MID 499 Ω
C61b194_ISO_MID 1 R3415 2 OUT_TIP 499 Ω
OUT_TIP 1 R3416 2 C61b194_FB 100 Ω
OUT_BUFFERED 1 C3401 2 C61b194_FB 1e-09 F
AGND S Tip Sleeve TN J1010 OUT_TIP Tip-normal WQP518MA
SELECTOR_COMMON 3 + C33d018_DRIVE 2 − U3403A 1 C33d018_DRIVE OPA4197IPWR
AGND 5 + CORE_PACK_OPA4197IPWR_0_IO_UNUSED2 6 − U3403B 7 CORE_PACK_OPA4197IPWR_0_IO_UNUSED2 OPA4197IPWR
AGND 10 + CORE_PACK_OPA4197IPWR_0_IO_UNUSED3 9 − U3403C 8 CORE_PACK_OPA4197IPWR_0_IO_UNUSED3 OPA4197IPWR
AGND 12 + CORE_PACK_OPA4197IPWR_0_IO_UNUSED4 13 − U3403D 14 CORE_PACK_OPA4197IPWR_0_IO_UNUSED4 OPA4197IPWR
+12V 4 V+ V− 11 −12V U3403E OPA4197IPWR
+12V 1 C3402 2 AGND 100 nF
−12V 1 C3403 2 AGND 100 nF
+12V 1 C3404 2 AGND 100 nF
−12V 1 C3405 2 AGND 100 nF
+12V 1 C3406 2 AGND 100 nF
−12V 1 C3407 2 AGND 100 nF
C025c3f_SENSE 1 R3417 2 AGND 5600 Ω
C025c3f_DRIVE 2 A K 1 C025c3f_LED_A D3403 BAS16GW-QX
C025c3f_SENSE 2 A K 1 C025c3f_LED_A D3404 BAS16GW-QX
C025c3f_LED_K 2 A K 1 C025c3f_DRIVE D3405 BAS16GW-QX
C025c3f_LED_K 2 A K 1 C025c3f_SENSE D3406 BAS16GW-QX
C025c3f_LED_K 1 D810 2 C025c3f_LED_A APHHS1005QWF/D
Cc6c62b_SENSE 1 R3418 2 AGND 5600 Ω
Cc6c62b_DRIVE 2 A K 1 Cc6c62b_LED_A D3408 BAS16GW-QX
Cc6c62b_SENSE 2 A K 1 Cc6c62b_LED_A D3409 BAS16GW-QX
Cc6c62b_LED_K 2 A K 1 Cc6c62b_DRIVE D3410 BAS16GW-QX
Cc6c62b_LED_K 2 A K 1 Cc6c62b_SENSE D3411 BAS16GW-QX
Cc6c62b_LED_K 1 D910 2 Cc6c62b_LED_A APHHS1005QWF/D
C7971de_SENSE 1 R3419 2 AGND 5600 Ω
C7971de_DRIVE 2 A K 1 C7971de_LED_A D3413 BAS16GW-QX
C7971de_SENSE 2 A K 1 C7971de_LED_A D3414 BAS16GW-QX
C7971de_LED_K 2 A K 1 C7971de_DRIVE D3415 BAS16GW-QX
C7971de_LED_K 2 A K 1 C7971de_SENSE D3416 BAS16GW-QX
C7971de_LED_K 1 D1010 2 C7971de_LED_A APHHS1005QWF/D
A_BUFFER 3 + C025c3f_SENSE 2 − U3404A 1 C025c3f_DRIVE OPA4196IDR
B_BUFFER 5 + Cc6c62b_SENSE 6 − U3404B 7 Cc6c62b_DRIVE OPA4196IDR
OUT_BUFFERED 10 + C7971de_SENSE 9 − U3404C 8 C7971de_DRIVE OPA4196IDR
AGND 12 + LED_PACK_OPA4196IDR_0_IO_UNUSED4 13 − U3404D 14 LED_PACK_OPA4196IDR_0_IO_UNUSED4 OPA4196IDR
+12V 4 V+ V− 11 −12V U3404E OPA4196IDR
+12V 1 C3408 2 AGND 100 nF
−12V 1 C3409 2 AGND 100 nF
A_SENSE 3 + A_BUFFER 2 − U3405A 1 A_BUFFER OPA4197IPWR
B_SENSE 5 + B_BUFFER 6 − U3405B 7 B_BUFFER OPA4197IPWR
SELECTOR_REMOTE 10 + C61b194_FB 9 − U3405C 8 OUT_BUFFERED OPA4197IPWR
AGND 12 + IO_EXTRA_5_IO_UNUSED4 13 − U3405D 14 IO_EXTRA_5_IO_UNUSED4 OPA4197IPWR
+12V 4 V+ V− 11 −12V U3405E OPA4197IPWR
+12V 1 C3410 2 AGND 100 nF
−12V 1 C3411 2 AGND 100 nF

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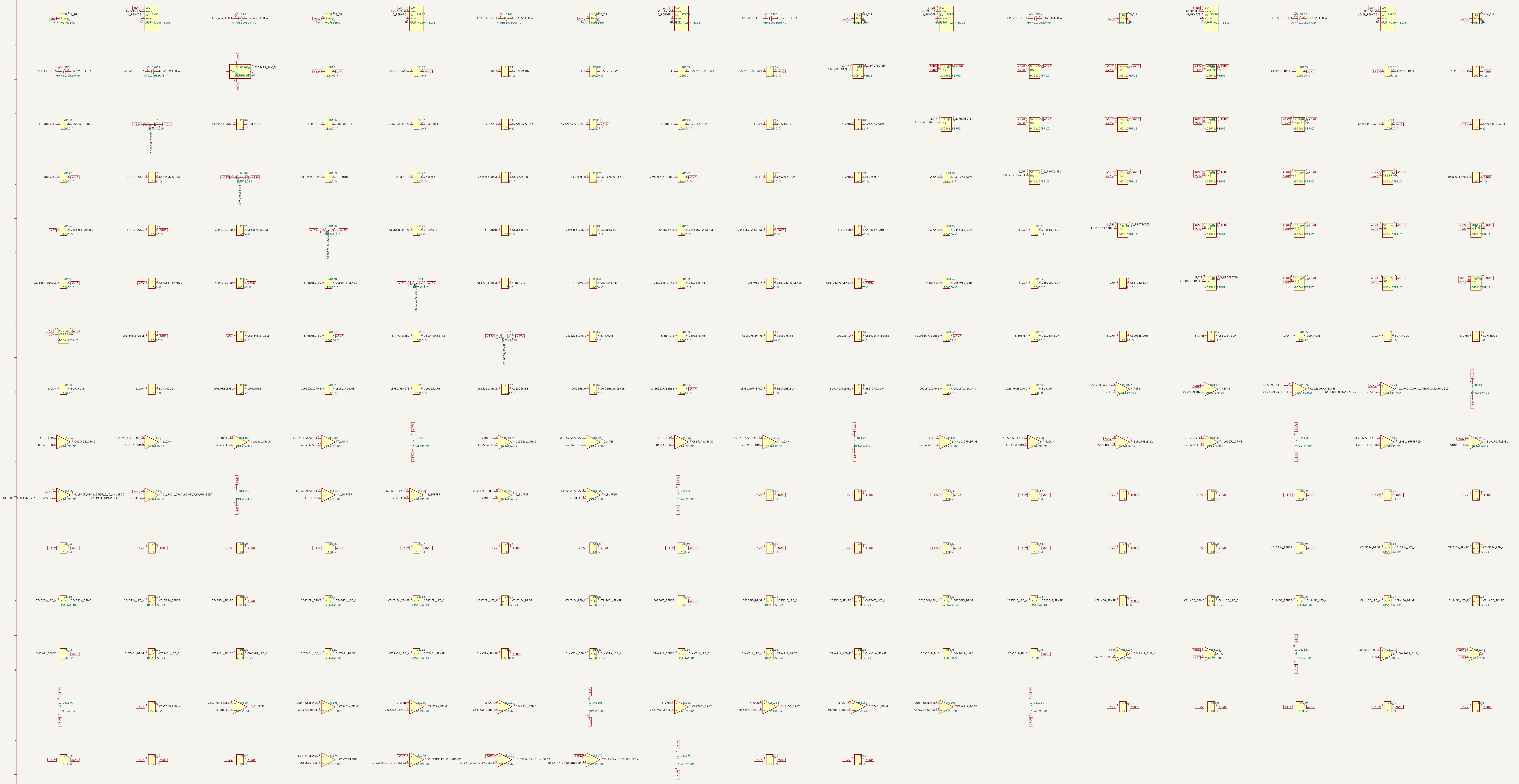

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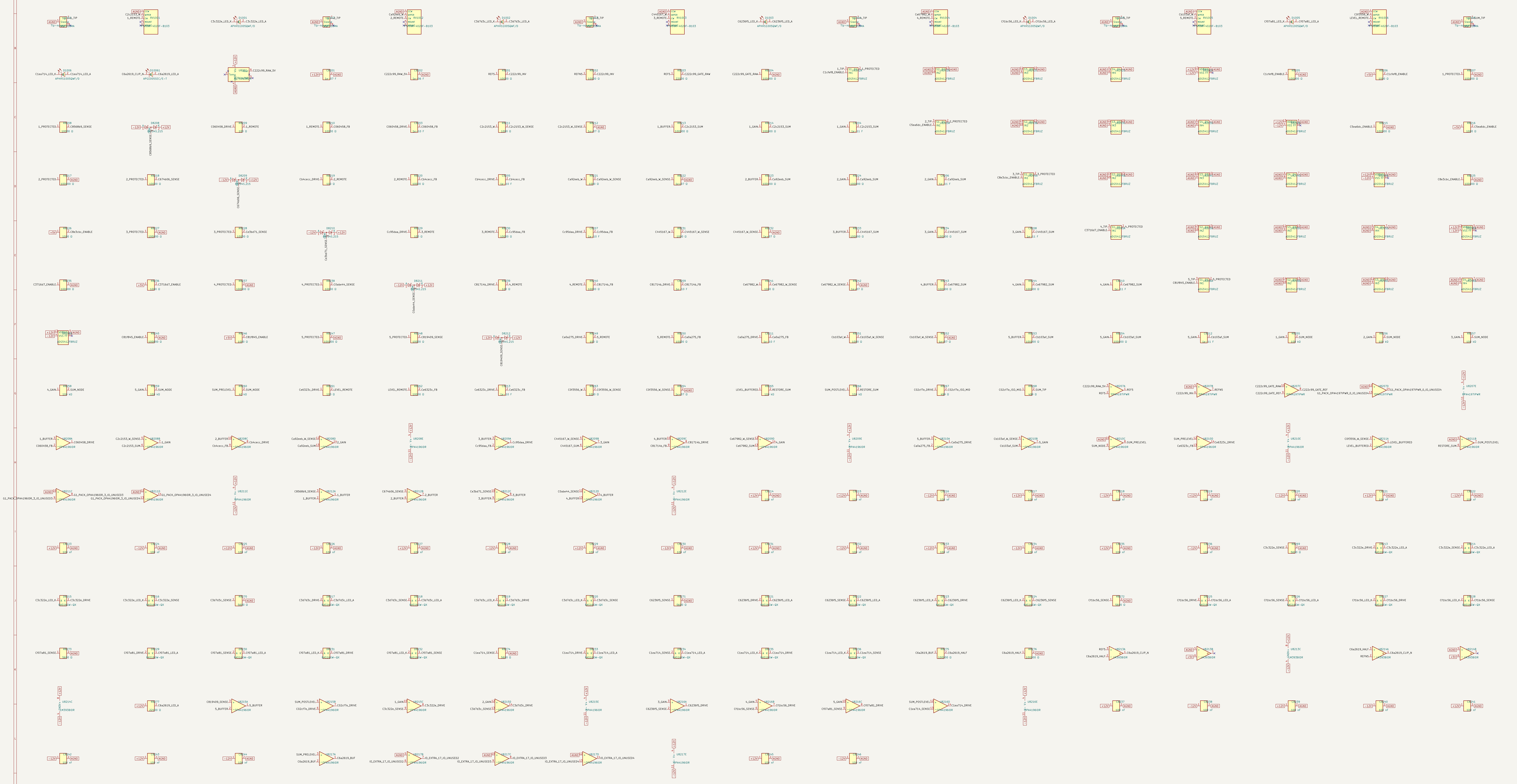

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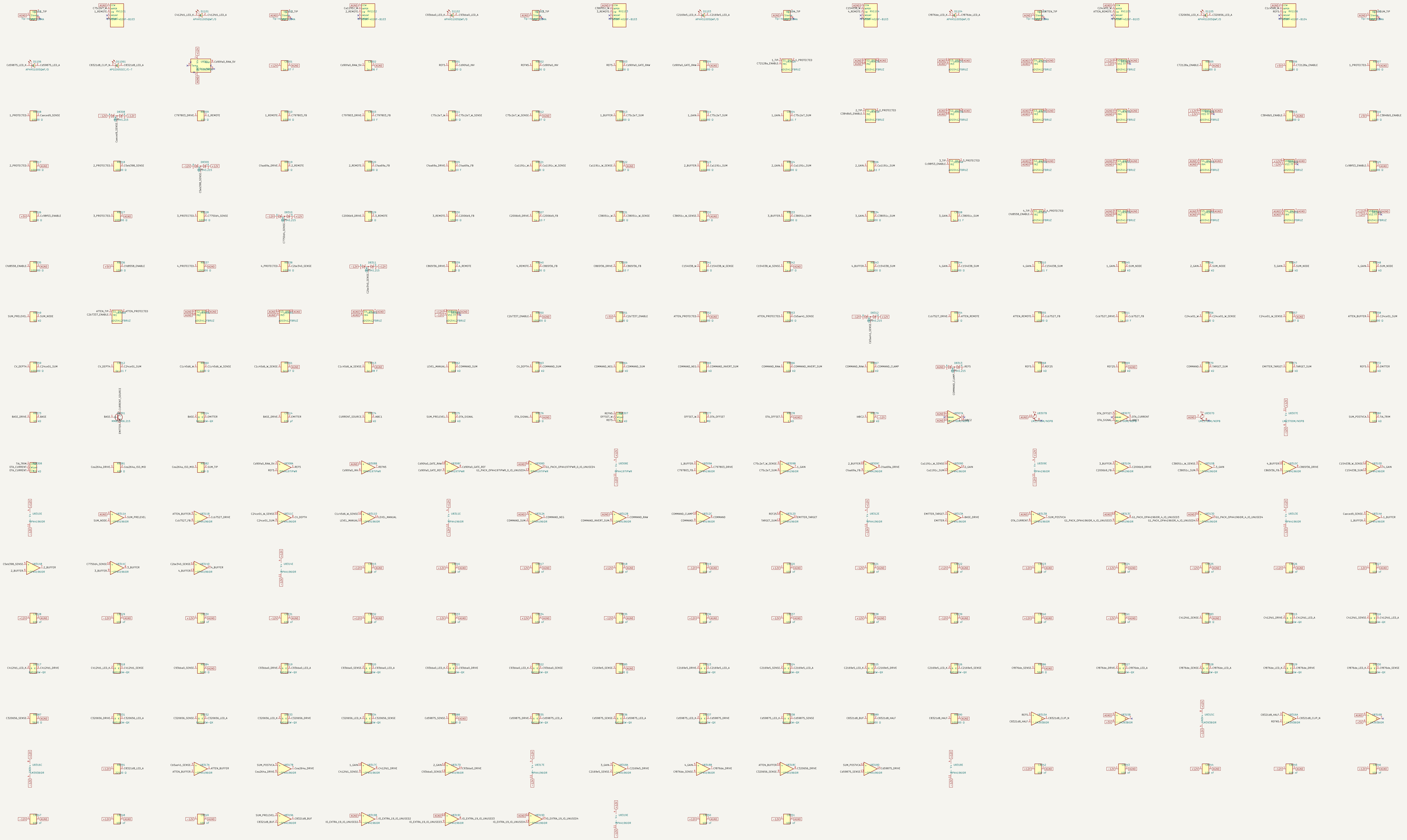

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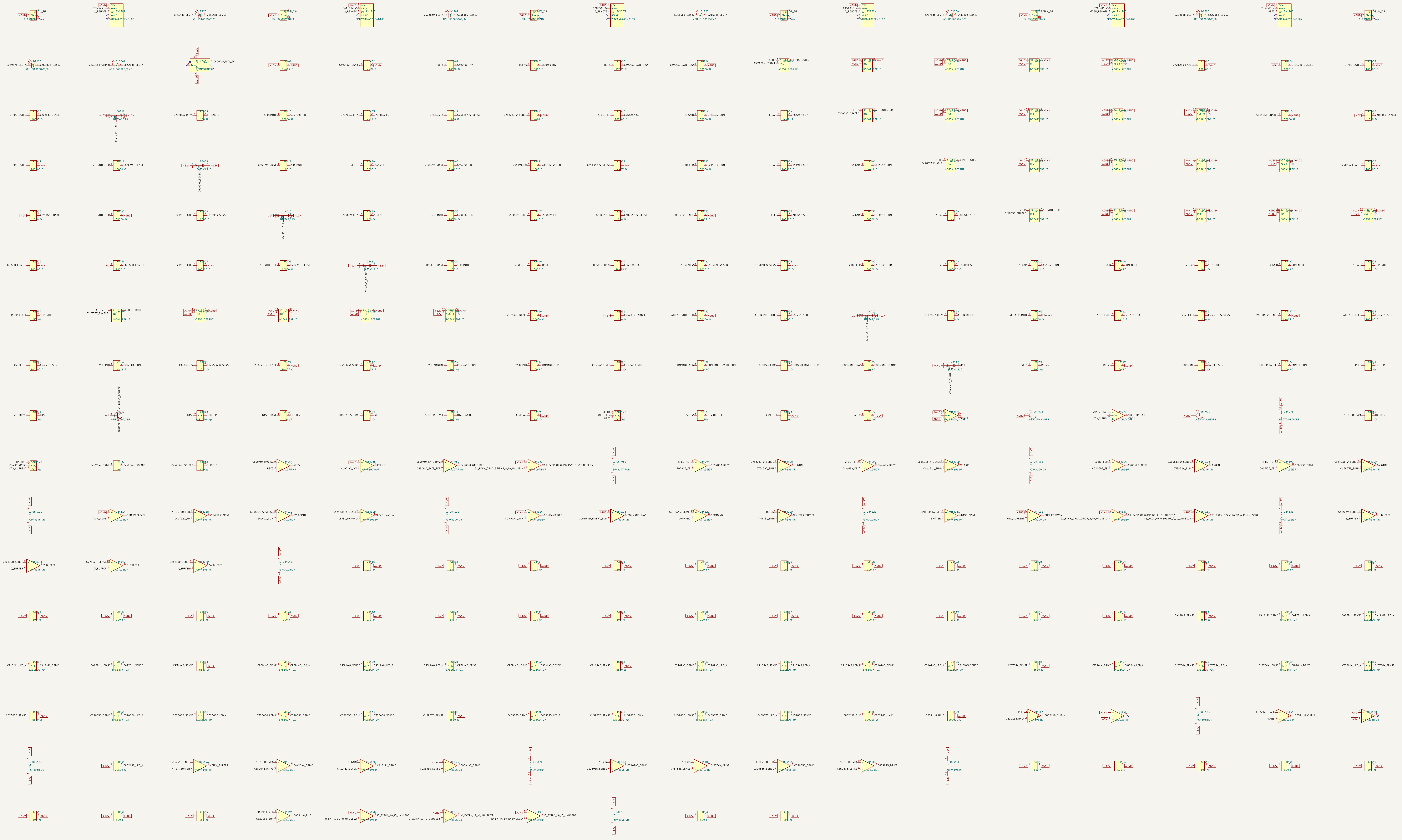

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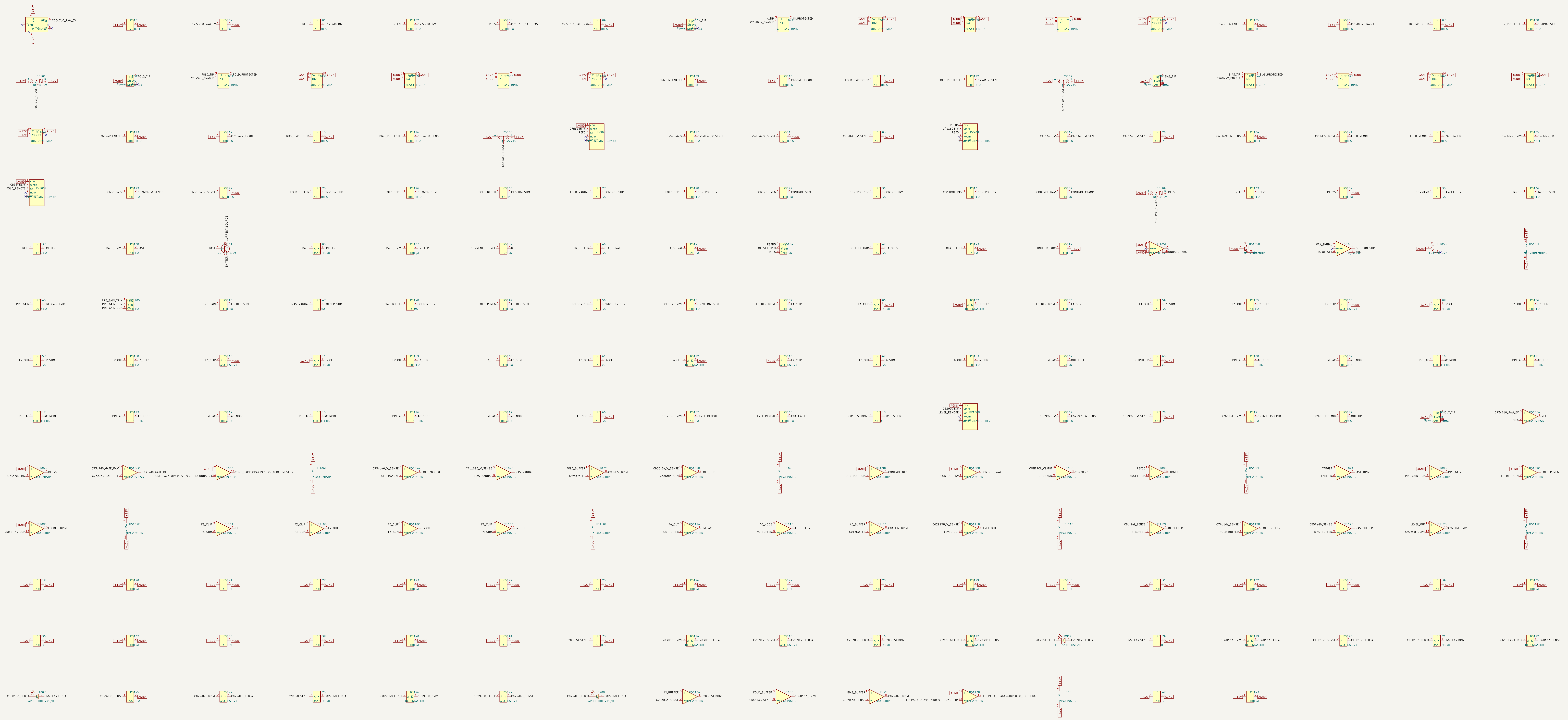

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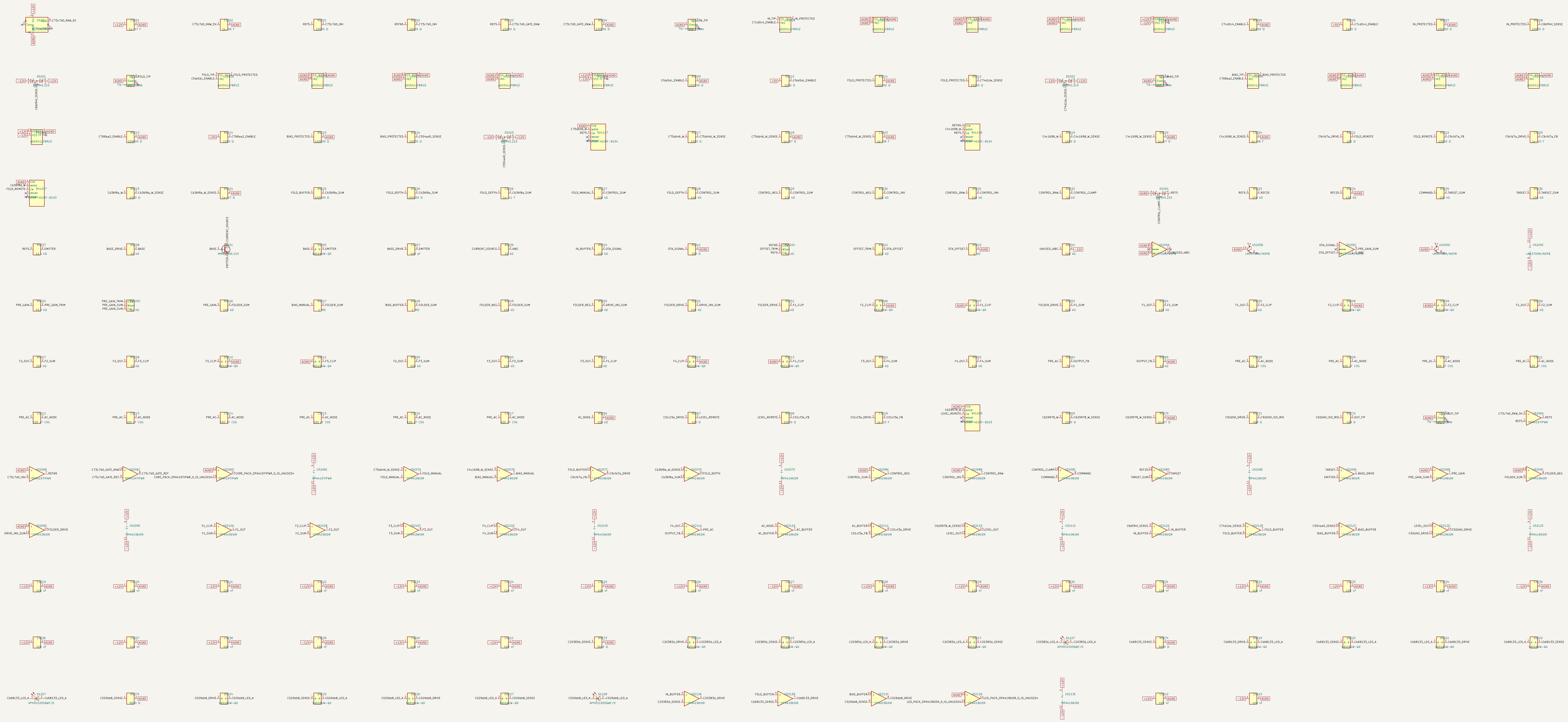

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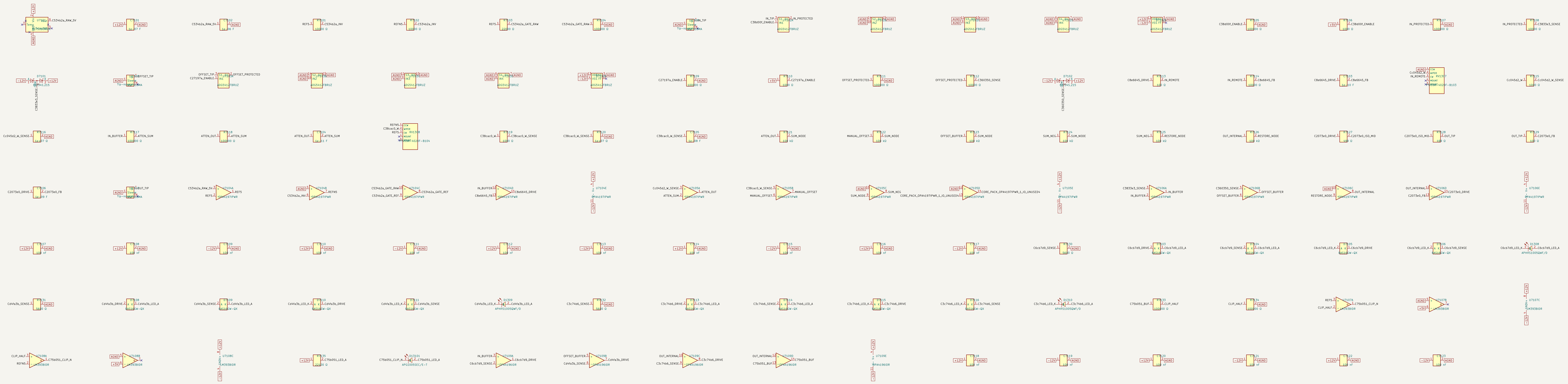

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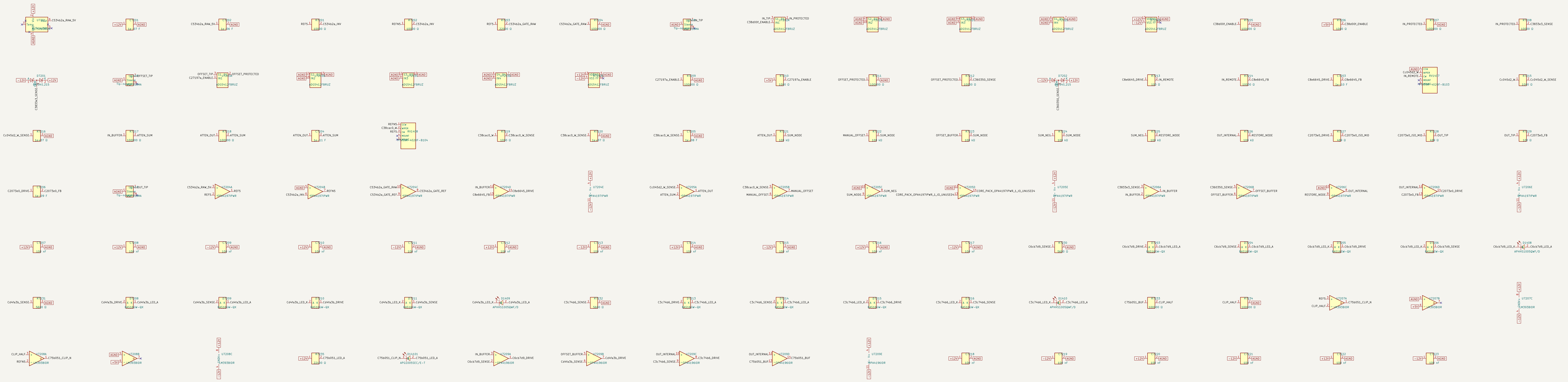
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C7201 1e-07 F +12V AGND
C7202 1e-06 F C534b2a_RAW_5V AGND
R7201 10000 Ω REF5 C534b2a_INV
R7202 10000 Ω REFN5 C534b2a_INV
R7203 22000 Ω REF5 C534b2a_GATE_RAW
R7204 100000 Ω C534b2a_GATE_RAW AGND
J1408 WQP518MA Tip Sleeve Tip-normal AGND IN_TIP
U7202A ADG5412FBRUZ S1 D1 IN1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7202B ADG5412FBRUZ S2 D2 IN2 AGND
U7202C ADG5412FBRUZ S3 D3 IN3 AGND
U7202D ADG5412FBRUZ S4 D4 IN4 AGND
U7202E ADG5412FBRUZ VDD VSS GND FF +12V -12V AGND
R7205 100000 Ω C38d00f_ENABLE AGND
R7206 1000 Ω +5V C38d00f_ENABLE
R7207 100000 Ω IN_PROTECTED AGND
R7208 10000 Ω IN_PROTECTED C3833e3_SENSE
D7201 BAT54S,215 A K COM -12V +12V C3833e3_SENSE
J1409 WQP518MA Tip Sleeve Tip-normal AGND OFFSET_TIP
U7203A ADG5412FBRUZ S1 D1 IN1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7203B ADG5412FBRUZ S2 D2 IN2 AGND
U7203C ADG5412FBRUZ S3 D3 IN3 AGND
U7203D ADG5412FBRUZ S4 D4 IN4 AGND
U7203E ADG5412FBRUZ VDD VSS GND FF +12V -12V AGND
R7209 100000 Ω C27197a_ENABLE AGND
R7210 1000 Ω +5V C27197a_ENABLE
R7211 100000 Ω OFFSET_PROTECTED AGND
R7212 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7202 BAT54S,215 A K COM -12V +12V C360350_SENSE
R7213 100 Ω C8e6645_DRIVE IN_REMOTE
R7214 10000 Ω IN_REMOTE C8e6645_FB
C7203 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1407 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7215 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7216 1e+07 Ω Cc045d2_W_SENSE AGND
R7217 100000 Ω IN_BUFFER ATTEN_SUM
R7218 100000 Ω ATTEN_OUT ATTEN_SUM
C7204 1e-11 F ATTEN_OUT ATTEN_SUM
RV1408 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7219 1000 Ω C38cac0_W C38cac0_W_SENSE
R7220 1e+07 Ω C38cac0_W_SENSE AGND
C7205 1e-08 F C38cac0_W_SENSE AGND
R7221 100 kΩ ATTEN_OUT SUM_NODE
R7222 100 kΩ MANUAL_OFFSET SUM_NODE
R7223 100 kΩ OFFSET_BUFFER SUM_NODE
R7224 100 kΩ SUM_NEG SUM_NODE
R7225 100 kΩ SUM_NEG RESTORE_NODE
R7226 100 kΩ OUT_INTERNAL RESTORE_NODE
R7227 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7228 499 Ω C2073e0_ISO_MID OUT_TIP
R7229 100 Ω OUT_TIP C2073e0_FB
C7206 1e-09 F C2073e0_DRIVE C2073e0_FB
J1410 WQP518MA Tip Sleeve Tip-normal AGND OUT_TIP
U7204A OPA4197IPWR C534b2a_RAW_5V REF5
U7204B OPA4197IPWR AGND C534b2a_INV REFN5
U7204C OPA4197IPWR C534b2a_GATE_RAW C534b2a_GATE_REF
U7204D OPA4197IPWR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7204E OPA4197IPWR V+ V- +12V -12V
U7205A OPA4197IPWR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7205B OPA4197IPWR C38cac0_W_SENSE MANUAL_OFFSET
U7205C OPA4197IPWR AGND SUM_NODE SUM_NEG
U7205D OPA4197IPWR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7205E OPA4197IPWR V+ V- +12V -12V
U7206A OPA4197IPWR C3833e3_SENSE IN_BUFFER
U7206B OPA4197IPWR C360350_SENSE OFFSET_BUFFER
U7206C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7206D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7206E OPA4197IPWR V+ V- +12V -12V
C7207 100 nF +12V AGND
C7208 100 nF +12V AGND
C7209 100 nF -12V AGND
C7210 100 nF +12V AGND
C7211 100 nF -12V AGND
C7212 100 nF +12V AGND
C7213 100 nF -12V AGND
C7214 100 nF +12V AGND
C7215 100 nF -12V AGND
C7216 100 nF +12V AGND
C7217 100 nF -12V AGND
R7230 5600 Ω C6cb7d9_SENSE AGND
D7203 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7204 BAS16GW-QX A K C6cb7d9_SENSE C6cb7d9_LED_A
D7205 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_DRIVE
D7206 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_SENSE
D1408 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7231 5600 Ω Cd4fa3b_SENSE AGND
D7208 BAS16GW-QX A K Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7209 BAS16GW-QX A K Cd4fa3b_SENSE Cd4fa3b_LED_A
D7210 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7211 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_SENSE
D1409 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7232 5600 Ω C3c74b6_SENSE AGND
D7213 BAS16GW-QX A K C3c74b6_DRIVE C3c74b6_LED_A
D7214 BAS16GW-QX A K C3c74b6_SENSE C3c74b6_LED_A
D7215 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_DRIVE
D7216 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_SENSE
D1410 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7233 100000 Ω C75b051_BUF CLIP_HALF
R7234 100000 Ω CLIP_HALF AGND
U7207A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7207B LM393BIDR AGND +5V
U7207C LM393BIDR GND V+ +12V -12V
U7208A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7208B LM393BIDR AGND +5V
U7208C LM393BIDR GND V+ +12V -12V
R7235 22100 Ω +12V C75b051_LED_A
D14101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7209A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7209B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7209C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7209D OPA4196IDR OUT_INTERNAL C75b051_BUF
U7209E OPA4196IDR V+ V- +12V -12V
C7218 100 nF +12V AGND
C7219 100 nF -12V AGND
C7220 100 nF +12V AGND
C7221 100 nF -12V AGND
C7222 100 nF +12V AGND
C7223 100 nF -12V AGND

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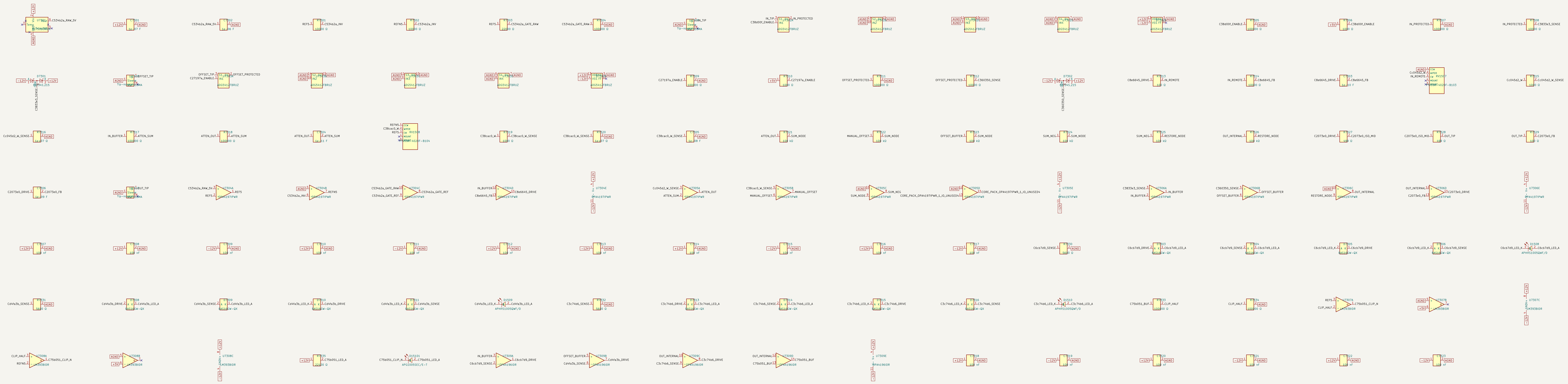

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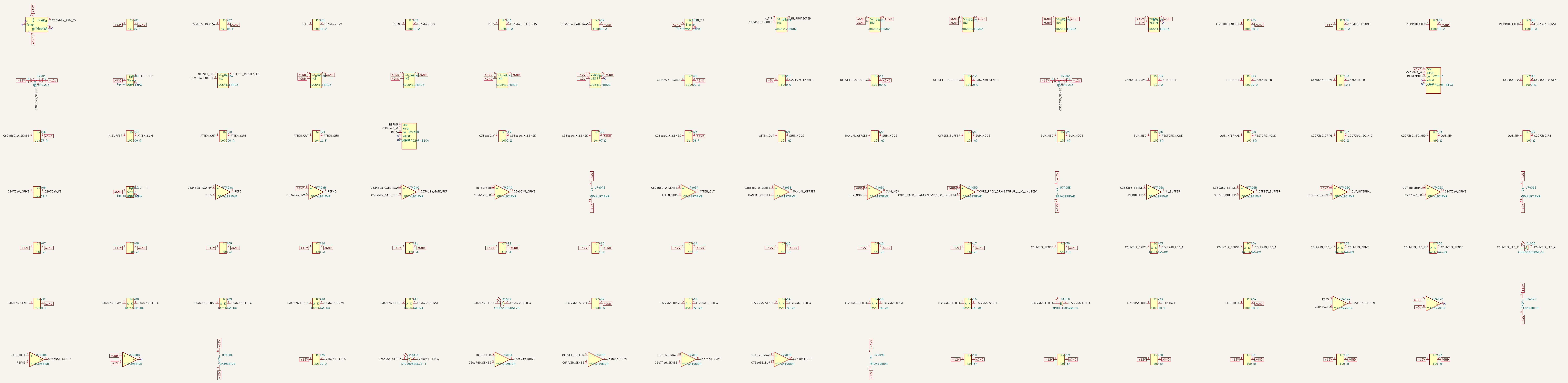

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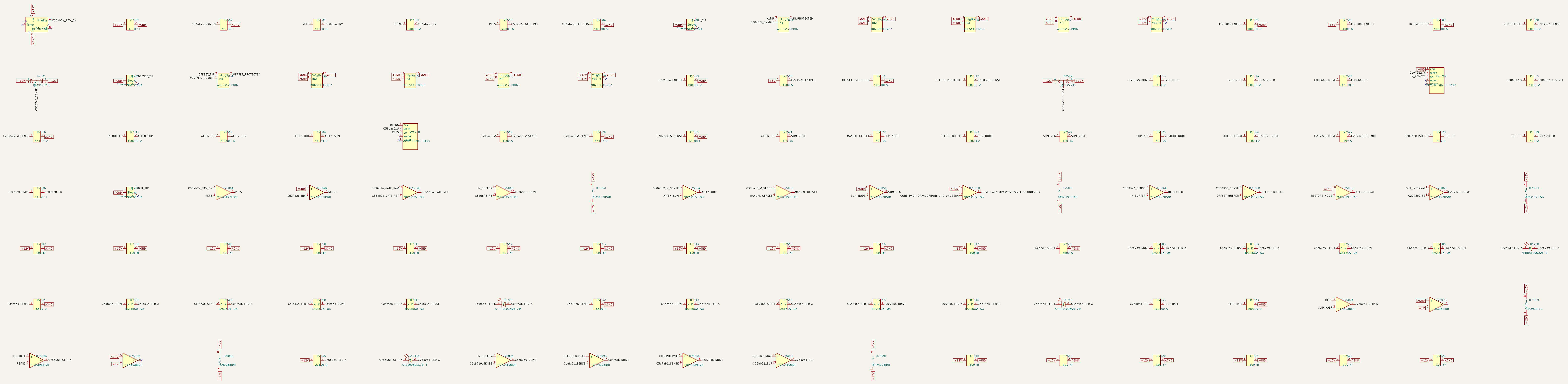

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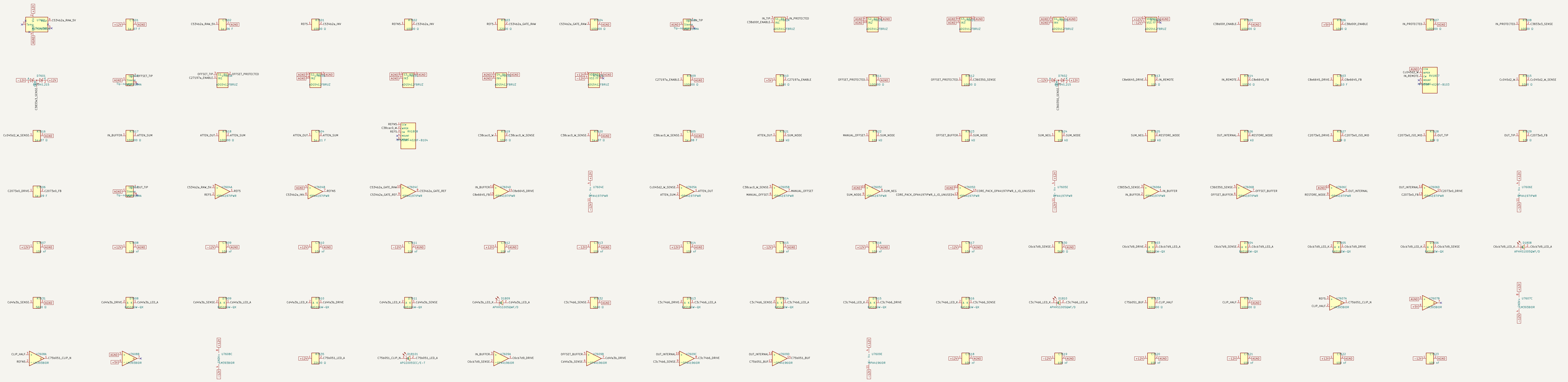
U7601 REF5050IDR Vin Vout Temp Trim/NR GND +12V AGND C534b2a_RAW_5V
C7601 +12V AGND 1e-07 F
C7602 C534b2a_RAW_5V AGND 1e-06 F
R7601 REF5 C534b2a_INV 10000 Ω
R7602 REFN5 C534b2a_INV 10000 Ω
R7603 REF5 C534b2a_GATE_RAW 22000 Ω
R7604 C534b2a_GATE_RAW AGND 100000 Ω
J1808 AGND Tip Sleeve TN Tip-normal IN_TIP WQP518MA
U7602A IN_TIP IN_PROTECTED C38d00f_ENABLE S1 D1 IN1 ADG5412FBRUZ
U7602B AGND S2 D2 IN2 ADG5412FBRUZ
U7602C AGND S3 D3 IN3 ADG5412FBRUZ
U7602D AGND S4 D4 IN4 ADG5412FBRUZ
U7602E +12V −12V VDD GND VSS FF AGND ADG5412FBRUZ
R7605 C38d00f_ENABLE AGND 100000 Ω
R7606 +5V C38d00f_ENABLE 1000 Ω
R7607 IN_PROTECTED AGND 100000 Ω
R7608 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7601 −12V A K +12V COM BAT54S,215 C3833e3_SENSE
J1809 AGND Tip Sleeve TN Tip-normal OFFSET_TIP WQP518MA
U7603A OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED S1 D1 IN1 ADG5412FBRUZ
U7603B AGND S2 D2 IN2 ADG5412FBRUZ
U7603C AGND S3 D3 IN3 ADG5412FBRUZ
U7603D AGND S4 D4 IN4 ADG5412FBRUZ
U7603E +12V −12V VDD GND VSS FF AGND ADG5412FBRUZ
R7609 C27197a_ENABLE AGND 100000 Ω
R7610 +5V C27197a_ENABLE 1000 Ω
R7611 OFFSET_PROTECTED AGND 100000 Ω
R7612 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7602 −12V A K +12V COM BAT54S,215 C360350_SENSE
R7613 C8e6645_DRIVE IN_REMOTE 100 Ω
R7614 IN_REMOTE C8e6645_FB 10000 Ω
C7603 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1807 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B103
R7615 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7616 Cc045d2_W_SENSE AGND 1e+07 Ω
R7617 IN_BUFFER ATTEN_SUM 100000 Ω
R7618 ATTEN_OUT ATTEN_SUM 100000 Ω
C7604 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1808 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B104
R7619 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7620 C38cac0_W_SENSE AGND 1e+07 Ω
C7605 C38cac0_W_SENSE AGND 1e-08 F
R7621 ATTEN_OUT SUM_NODE 100 kΩ
R7622 MANUAL_OFFSET SUM_NODE 100 kΩ
R7623 OFFSET_BUFFER SUM_NODE 100 kΩ
R7624 SUM_NEG SUM_NODE 100 kΩ
R7625 SUM_NEG RESTORE_NODE 100 kΩ
R7626 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7627 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7628 C2073e0_ISO_MID OUT_TIP 499 Ω
R7629 OUT_TIP C2073e0_FB 100 Ω
C7606 C2073e0_DRIVE C2073e0_FB 1e-09 F
J1810 AGND Tip Sleeve TN Tip-normal OUT_TIP WQP518MA
U7604A C534b2a_RAW_5V REF5 REF5 OPA4197IPWR
U7604B AGND C534b2a_INV REFN5 OPA4197IPWR
U7604C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IPWR
U7604D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IPWR
U7604E +12V −12V V+ V− OPA4197IPWR
U7605A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IPWR
U7605B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IPWR
U7605C AGND SUM_NODE SUM_NEG OPA4197IPWR
U7605D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IPWR
U7605E +12V −12V V+ V− OPA4197IPWR
U7606A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7606B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7606C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7606D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7606E +12V −12V V+ V− OPA4197IPWR
C7607 +12V AGND 100 nF
C7608 +12V AGND 100 nF
C7609 −12V AGND 100 nF
C7610 +12V AGND 100 nF
C7611 −12V AGND 100 nF
C7612 +12V AGND 100 nF
C7613 −12V AGND 100 nF
C7614 +12V AGND 100 nF
C7615 −12V AGND 100 nF
C7616 +12V AGND 100 nF
C7617 −12V AGND 100 nF
R7630 C6cb7d9_SENSE AGND 5600 Ω
D7603 C6cb7d9_DRIVE C6cb7d9_LED_A A K BAS16GW-QX
D7604 C6cb7d9_SENSE C6cb7d9_LED_A A K BAS16GW-QX
D7605 C6cb7d9_LED_K C6cb7d9_DRIVE A K BAS16GW-QX
D7606 C6cb7d9_LED_K C6cb7d9_SENSE A K BAS16GW-QX
D1808 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7631 Cd4fa3b_SENSE AGND 5600 Ω
D7608 Cd4fa3b_DRIVE Cd4fa3b_LED_A A K BAS16GW-QX
D7609 Cd4fa3b_SENSE Cd4fa3b_LED_A A K BAS16GW-QX
D7610 Cd4fa3b_LED_K Cd4fa3b_DRIVE A K BAS16GW-QX
D7611 Cd4fa3b_LED_K Cd4fa3b_SENSE A K BAS16GW-QX
D1809 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7632 C3c74b6_SENSE AGND 5600 Ω
D7613 C3c74b6_DRIVE C3c74b6_LED_A A K BAS16GW-QX
D7614 C3c74b6_SENSE C3c74b6_LED_A A K BAS16GW-QX
D7615 C3c74b6_LED_K C3c74b6_DRIVE A K BAS16GW-QX
D7616 C3c74b6_LED_K C3c74b6_SENSE A K BAS16GW-QX
D1810 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7633 C75b051_BUF CLIP_HALF 100000 Ω
R7634 CLIP_HALF AGND 100000 Ω
U7607A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7607B AGND +5V LM393BIDR
U7607C +12V −12V V+ GND LM393BIDR
U7608A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7608B AGND +5V LM393BIDR
U7608C +12V −12V V+ GND LM393BIDR
R7635 +12V C75b051_LED_A 22100 Ω
D18101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7609A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7609B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7609C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7609D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7609E +12V −12V V+ V− OPA4196IDR
C7618 +12V AGND 100 nF
C7619 −12V AGND 100 nF
C7620 +12V AGND 100 nF
C7621 −12V AGND 100 nF
C7622 +12V AGND 100 nF
C7623 −12V AGND 100 nF

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